Wajih Habrah

Biomedical Engineer | Robotics, Digital Twins and Medical Technology

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Portfolio: https://wajihhabrah.github.io

Kapellgången 3, 41131 Gothenburg, Sweden

Education

Chalmers University of Technology Gothenburg, Sweden

Master's Degree in Biomedical Engineering *Sep. 2023 – Jun. 2026*

- **Thesis:** Robotic Phantom Knee and Digital Twin Platform for Early-Stage Exoskeleton Validation
  Official publication: https://hdl.handle.net/20.500.12380/311285
- **Selected coursework:** Deep Machine Learning, Applied Digital Health, Image Analysis, MRI, Rehabilitation Engineering, Modelling and Simulation in Biomedical Engineering, Applied Signal Processing, Advanced Topics in Biomedical Engineering.

Damascus University Damascus, Syria

Bachelor's Degree in Biomedical Engineering *2007 – 2013*

- **Selected coursework:** Biomechanics, Biomedical Control, Automatic Control, Modelling and Simulation, Signal Processing, Medical Electronics & Bio-measurements, Medical Devices, Microprocessors, Artificial Extremities, Image Systems & Image Processing.

Work Experience

Biomedical Engineer (5 years) Jan. 2018 – Dec. 2021 full-time; Jan. 2022 – Present hourly/consulting

Southern hospital of Stockholm - Södersjukhuset *Stockholm, Sweden*

- Performed maintenance, calibration and troubleshooting of ICU systems, C-arms and critical diagnostic medical equipment.
- Developed and improved preventive maintenance procedures, including C-arm service routines, to increase operational reliability and safety.
- Produced and maintained technical documentation and supported clinical staff in equipment use, incident reporting and safety follow-ups.

Selected Projects

Robotic Phantom Knee and Digital Twin | *ROS 2, Unity, Python, C#, Raspberry Pi, Sensors, Actuators* 2025 – 2026

- Designed and implemented a robotic knee testbed for early-stage exoskeleton validation, combining physical hardware with a real-time Unity digital twin.
- Developed bidirectional data flows between the physical system and the digital twin for real-time visualization, debugging and controller prototyping.
- Integrated variable-stiffness antagonistic actuation, pneumatic soft-tissue simulation and sensing from IMUs, pressure sensors, temperature sensors and encoders.
- Developed ROS 2 nodes for robotic motor control, pneumatic actuation, sensor integration, parameter management, logging and safety logic such as E-stop and air-release behavior.
- Built the Unity side of the system using ROS-TCP communication for interactive visualization and hardware-in-the-loop experimentation.

Personal 3D and Interactive Systems R&D | *Unity, Blender, Firebase, 3D Modelling* 2019 – Present

- Explored interactive 3D systems using Unity and Blender, including asset workflows, real-time interaction and backend prototypes for dynamic configuration.
- Created 3D models and prepared assets for visualization, texture baking, low-poly modelling and 3D-printing workflows.

Technical Skills

Robotics & Control: ROS 2, rclpy, real-time control, closed-loop control, cooperative actuation, variable-stiffness actuation, safety logic
Digital Twins & Simulation: Unity, ROS-TCP-Connector, hardware-in-the-loop visualization, Blender, 3D-printing preparation
Programming: Python, C#, C++, MATLAB
Sensors & Hardware: IMUs, pressure sensors, temperature sensors, encoders, Raspberry Pi, microcontrollers, pneumatic systems, medical equipment
Tools: Git, GitHub, GitLab, Unity Version Control, Microsoft Office

Languages

Arabic: Native
English: Fluent
Swedish: Fluent

Additional

Nationality: Dual nationality, Syrian and Swedish
Driving License: Swedish class B, since 2017
Interests: Basketball, portrait drawing, sculpture, vehicle troubleshooting and repair
References: Available upon request